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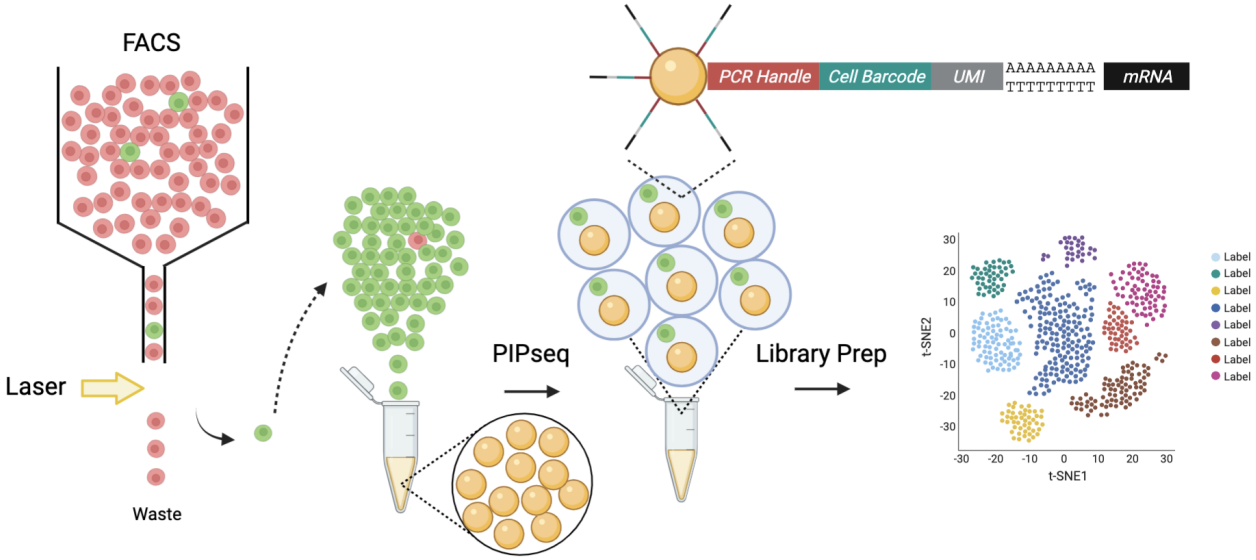
FACs for PIPseq

Demonstrated Protocol

Doc ID: **FB0002199**
Revision: **3**
State: **Release**

Overview

This Demonstrated Protocol describes the best practice for enriching rare cell types with Fluorescence-Activated Cell Sorting (FACS) for direct input into the PIPseq™ Single Cell 3' Gene Expression workflow. This protocol is recommended for rare cell types in the 0.1-0.01% range, which will allow for sufficient enrichment into PIPseq with minimal added time to the workflow to maximize cell health. Users are welcome to FACS enrich for cell types rarer than 0.01%, but longer sorting times may negatively impact the quality of the results.



References

Doc ID	Title
FB0001026	PIPseq T2 v3.0 Single Cell RNA Kit User Guide

Materials and Methods

Required Third Party Reagents and Consumables

Reagent	Supplier
Cell Counting Materials using laboratory's preferred method (e.g. AO/PI)	Preferred Vendor
Sterile Tips 20 µL, filtered, low retention	Rainin; Cat# 30389226 VWR;76322-528
Sterile Tips 200 µL, filtered, low retention	Rainin; Cat# 30389240 VWR; Cat#76322-150
Sterile Tips 1000 µL, filtered, low retention	Rainin; Cat# 30389213 VWR; Cat#76322-154
15 mL conical tube, sterile and nuclease-free	General laboratory supplier
Flowmi Tip Strainer, 40 micron	VWR H13680-0040

Required Equipment

Description	Supplier
FACS instrument	Preferred equipment supplier
Hemocytometer or automated cell counter	General laboratory supplier

Protocol

Cell Preparation

The FACS for PIPseq™ Single Cell 3' Gene Expression protocol requires a suspension of viable single cells as input which may be derived from cell culture, dissociated tissues, cell sorting or other isolation methods. Users should minimize the presence of dead or aggregated cells to ensure the highest quality data. Preparation of single cells from tissues or fragile cell types may require additional dissociation or cell est.

1. Stain cells of interest according to the preferred dye or antibody protocol optimized for FACS and save the excess unstained cells
2. Centrifuge both cell types at 200 x g for 5 min to pellet cells.

   The cell pellet may form on the side or the bottom of the tube depending on the rotor type. Swinging bucket rotors are recommended for pelleting. Centrifugation speed depends on cell type and size. Generally, up to 300g x 5 min can be used safely for smaller cell types. The goal is to form a loosely packed pellet with little to no cells in the supernatant. To this end, when working with new cell types it is important to examine the supernatant and monitor the difficulty of resuspension (such as pellet packed too tight) until you find the minimum speed and time required for optimal pelleting.
3. Aspirate as much of the supernatant as possible without disturbing the cell pellet.
4. Resuspend in 1 mL of 0.1% BSA in DPBS.
5. Using a **wide-bore P1000** tip, withdraw the 1 mL cell suspension and add a 40 micron FlowMi tip strainer onto the end of the tip
6. Gently dispense the cell suspension through the tip strainer into a clean 1.5 mL tube.
7. Determine cell concentration using the laboratory's preferred method (e.g. trypan blue staining on an automated cell counter).
8. Refer to Table 1 and prepare an appropriate dilution of the cell suspension on ice with 0.1% BSA in DPBS according to the FACS instrument specifications.
9. Thaw one PIP tube per reaction on ice for at least 15 minutes. After the PIP tubes have fully thawed, centrifuge them for ~5 seconds on a benchtop minifuge to remove air bubbles.
10. Prepare a 0.5 mL PCR tube with 15 µL of appropriate cell culture media to serve as a counting control for the FACS instrument. Users will sort cells into this tube and count them to validate the number of cells sorted into the PIP tube.
11. Assemble the following reagents on ice and transfer to the FACS instrument
    a. Unstained Cells in 0.1% BSA in DPBS
    b. Stained cells in 0.1% BSA in DPBS
    c. Thawed PIPs tube, 1 per reaction
    d. A 0.5 mL PCR tube adaptor (Necessary to hold PIP tube in FACS instrument)

e. Clean 0.5 mL PCR tubes

f. Counting control: 0.5 mL PCR tube with 15 µL of appropriate media

Instrument Specifications					
Sort Set–ups					
Nozzle (µM)	70	85	100		
Pressure (PSI)	70 (high)	35 (medium)	30 (medium)	20 (low)	
Cell Type	Splenocytes Thymocytes PBMC whole blood bacteria yeast bone marrow	B/T cell lines Embryonic Stem Cells Transduced/Transfected spleen, thymus bone marrow (delicate) cells	Embryonic stem cells Trophoblast stem cells Cultured primary cells	Endothelial cells All adherent cell lines Newly transfected cells Large delicate primary cells*	
Cell Concentration**	40 million/mL	10–15 million/mL	8 million/mL	3–5 million/mL (bulk sorts)	1 million/mL (plate sorts)
Threshold Rate	18–22 x 10^3 cell/sec	~8,000 events/sec	~6,000 events/sec	~3,000 events/sec	~500–1,000 events/sec
Optimal Injection Speed***	20–35 µL/min				

Table 1. Instrument specifications for BD Aria instruments

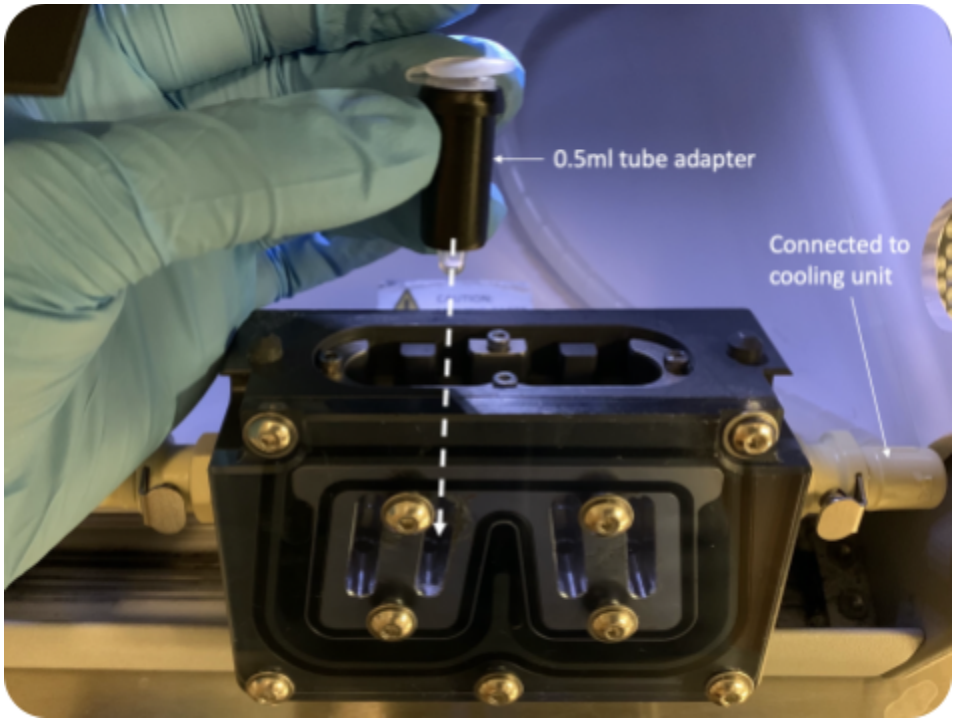
**Cell Concentration: values given are a maximum concentration and concentrations higher than this will require dilution and increased sorting time.

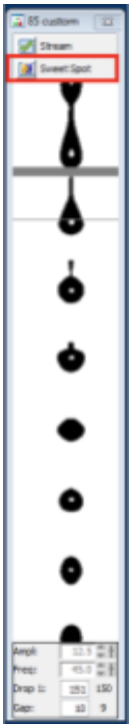
***Injection speed is dependent on actual cell concentration. At 30 µL/min the instrument will finish 1mL of sample in 35 minutes.

FACS with the BD Aria 2, 3 or Fusion

This Demonstrated Protocol is specific to FACS with the BD Aria 3 instrument. Users may adapt the protocol as necessary to the specifics of their preferred FACS instrument.

1. Turn the cooling unit on and connect it to the 4-way 1.5 mL tube holder. It takes approximately 30 minutes to cool down to 4˚C. Insert the 0.5 mL tube adapter into one of the slots.





2. Turn the stream on and wait for the stream to stabilize. Turn on Sweetspot.

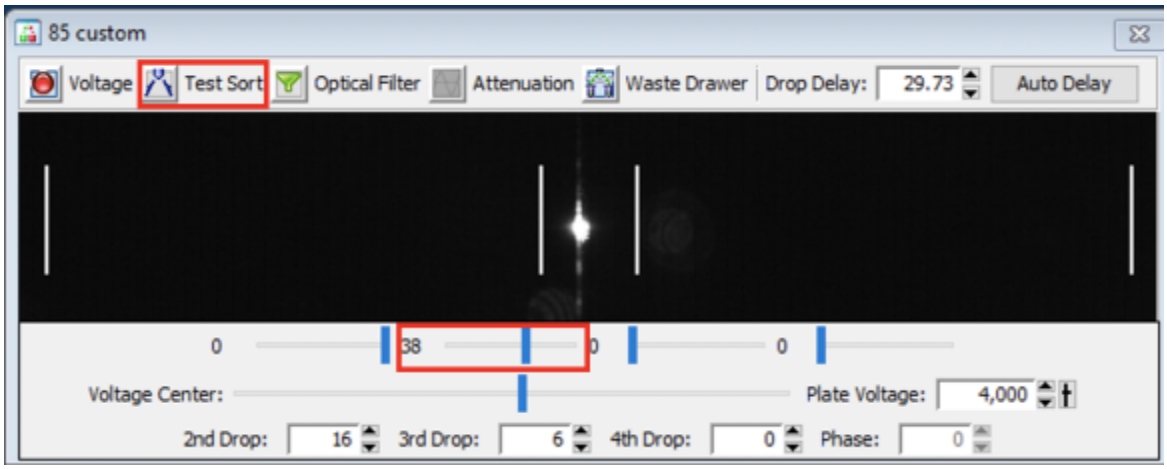
   Note: Sweetspot is a function in the FACS program that tracks drift in the Gap and Drop 1 measurements. It automatically adjusts amplitude to return their value to the optimal range. It will automatically stop the sorting when the values are out of range and resume once the values are back in the specified range.

3. Create new Experiment
4. Load unstained cells
   a. Adjust FSC Area Scaling in Cytometer->Laser
   b. Set Parameters in use in Cytometer -> Parameters and set voltages
   c. Create gates
5. Load stained cells
   a. Create New Tube -> Rename
   b. Start with lowest flow rate = 1
   c. Align sort stream
      i. Insert an empty 0.5 mL tube (lid closed) into the 0.5 mL tube adapter.
      ii. In the Side Stream window, click the "Voltage" button to turn on plate voltage (turns red).
      iii. Click the "Waste Drawer" button to open the drawer.
      iv. Test Sort: Double click the button to quickly sort 1 drop and close it

v. Check the lid of the 0.5 mL tube to see if the droplet is located at the center. If not, adjust the slider bars. Redo Step 4-5 until the droplet is centered.

vi. Open the lid and **Test Sort** one droplet into the tube. Check if the droplet is at the bottom of the tube. If the droplet is on the wall, adjust the **slider bars** and redo this step.

vii. When the sort stream is optimized, close the Waste Drawer and turn off the Voltage.

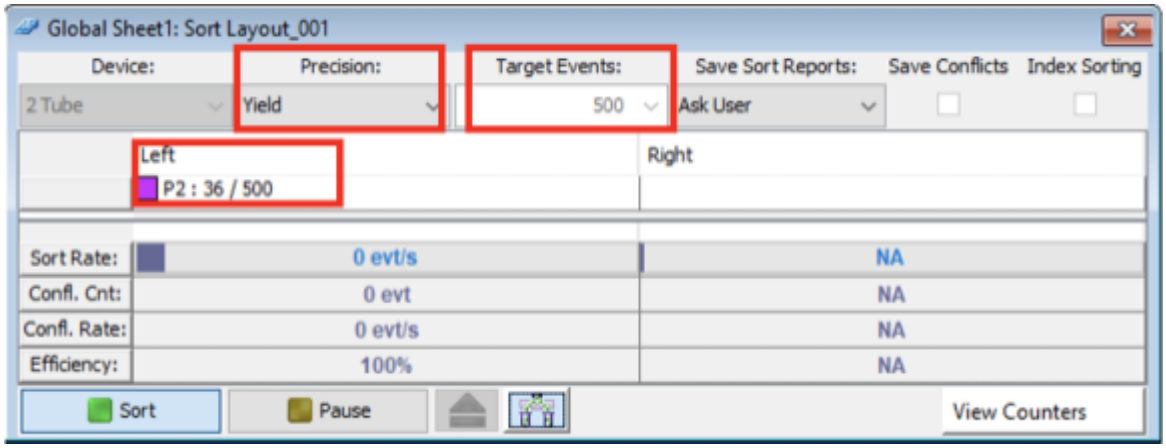


d. Sort 100 cells into media for cell count control

i. Load medium-containing 0.5 mL tube (lid open) into the 0.5 mL tube adapter.

ii. In the Sort Layout window, select "**yield**" mode. Select the **target gate** and input 100. Start Sort.

iii. Once sorting is complete, unload the collection tube and spin on a minifuge for ~2 s. Keep the tube on ice.



e. Sort 2500 cells into PIPtube

i. Start Acquire

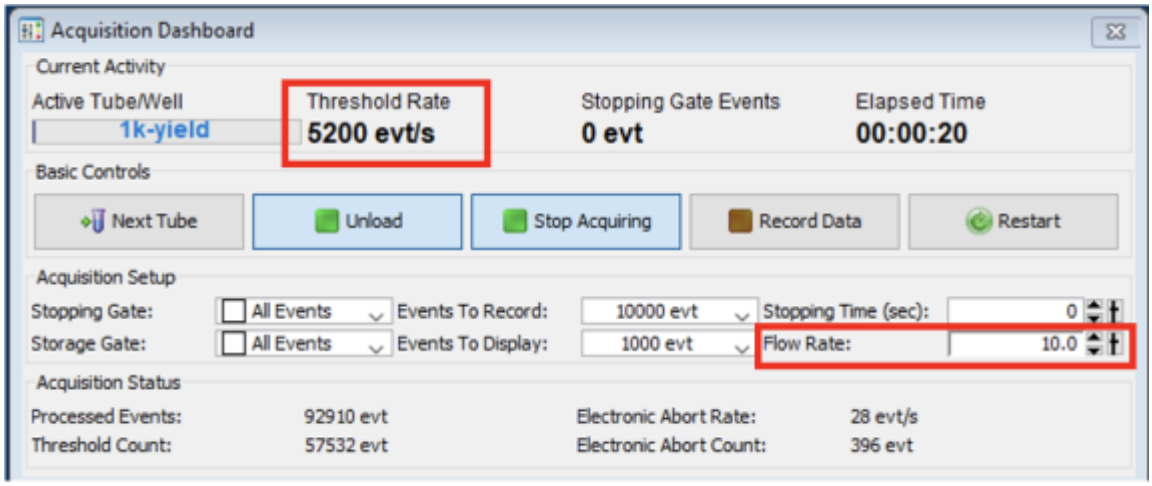
ii. Load PIP tube (lid open) into the 0.5 mL tube adapter

iii. In the Sort Layout window, select the target gate and input 2500. Start Sort.

iv. Increase **Flow Rate** until **Threshold Rate** is close to 8000 events/s when using 85 µm nozzle (see table 1)

v. Once sorting is complete (or time has reached 30 min or sample volume has run out), unload the PIP tube and minifuge spin for ~2 s. Keep the PIP tube on ice.

vi. Unload sample tube and do the cleaning steps for FACS.



After FACS

1. Count cells from the Dummy tube (0.5 mL tube with 15 µL media). Load the entire volume into a 20 µL hemocytometer chamber. Count in both fluorescent (target cells) and bright-field (mis-sort) channels. Estimate the recovery rate and input cell number for the PIP tube.
    a. Recovery Rate = [(Total number of cells in the collection tube × Purity) / Number of target cells reported to have been sorted by the instrument] × 100
    b. Input cell number for PIP tube = 2500 (or the actual sort number if 2500 is not reached) x Recovery Rate
2. If the input cell number for the PIP tube is acceptable, proceed to step 3.
3. Calculate the input volume into the PIP tube using the Sort Droplet Volume Calculator spreadsheet. Fill the volume to 5 µL by adding Cell Suspension Buffer into the PIP tube.
4. Proceed to PIPseq protocol from step Cell Capture and Lysis Step 6
    a. For this mixing step, make sure to layer particles to the top to ensure the cells are completely mixed in the PIP layer.

Alternatively, when not FACS sorting directly into PIPs we recommend you count your cells post-FACS, and proceed to Step 6 of General Cell Preparation in *FB0001026 PIPseq T2 v3.0 Single Cell RNA Kit User Guide.*

Document Revision Summary

Doc ID: FB0002199
Revision: 3
Revision date: Jan 2023

General Changes:

- Changes to improve readability

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